

increase in hedging/deference (63.2%, $z = 3.36$, $p < 0.001$).

These results reveal domain-specific sycophancy profiles. In moral judgment scenarios (AITA), personalization increases emotional support and indirectness but does not systematically affect how models frame responsibility or assign blame. In open-ended personal advice (OEQ), personalization increases emotional support and linguistic indirectness while decreasing acceptance of users' problem framing.

3.2. Feedback Sycophancy: SYCON-Debate

Personalization significantly increases AI openness to user opinion statements in debate contexts, relative to the generic treatment condition (Mean = 70.4%, range: 55.2%-85.0%) (Figure 2a). Challenge statements elicit stronger personalization effects (Mean = 72.6%, range: 61.1%-85.0%) compared to agreement statements (Mean = 68.1%, range: 55.2%-82.1%). Five models show significant effects for agreement statements (z -values: 3.62-5.46, $ps < 0.001$) and eight models for challenge statements (z -values: 2.45-5.95, $ps < 0.05$), with four models showing significant effects for both response types. Claude Haiku 4.5 (agree: 57.6%, $z = 1.25$, $p = .21$), Grok 3 Mini (agree: 56.4%, $z = 1.05$, $p = .29$), Llama 3.1 8B (agree: 55.2%, $z = 0.85$, $p = .39$), and Mistral Small 3.2 24B (agree: 57.6%, $z = 1.25$, $p = .21$) show no significant effect for agreement statements, while only Llama 3.3 70B shows a marginal non-significant effect for challenge statements (61.1%, $z = 1.85$, $p = .06$). Beyond the overall increase in openness, we found that LLMs actively calibrate their degree of sycophancy based on inferred user preferences. Personas rated as preferring validation were significantly more likely to receive more open personalized responses, for both challenge ($\beta = 0.57$, $SE = 0.18$, $p = .002$) and agreement statements ($\beta = 0.57$, $SE = 0.24$, $p = .019$).

To isolate the effect of personalization, we measured rates of sycophancy under four conditions (neutral text, demographics only, third person, and first person personalized) for SYCON-Debate using Qwen 3 235B as a representative model (Figure 2b). All forms of personalized information increase AI openness to user opinion compared to neutral text (agree: 46.4%; challenge: 50.0%). Demographics-only profiles (agree: 74.3%, $z = 4.13$, $p < 0.001$; challenge: 83.5%, $z = 5.70$, $p < 0.001$), third-person profiles (agree: 67.6%, $z = 2.99$, $p = .003$; challenge: 82.5%, $z = 5.54$, $p < 0.001$), and first-person personalized profiles (agree: 78.5%, $z = 4.86$, $p < 0.001$; challenge: 84.5%, $z = 5.86$, $p < 0.001$) all produce significantly more open responses in the personalized condition than expected if there was no difference between treatment conditions. The neutral text condition shows no significant difference between personalized and generic responses for either agreement (46.4%, $z = -0.61$, $p = .54$) or challenge statements (50.0%, $z = -0.01$, $p = 1.00$). Sycophantic behavior is specifically driven by the presence

of personalized information rather than by the addition of any contextual text. Even minimal demographic information is sufficient to trigger substantial increases in openness to user opinion, though richer first-person profiles produce slightly stronger effects.

3.3. Persistent Rebuttals over Multi-Turn Conversations Significantly Increase Sycophancy

After establishing that personalization increases sycophantic language in one-turn responses, we evaluate the effect of repeated challenges, combined with personalization, in multi-turn conversations (Figure 3a). Across treatment conditions, over 10 turns of persistent user rebuttals, AI becomes significantly more likely to adopt the user's viewpoint, indicated by an increase in debate stance flip rate ($\beta = 0.50$, $SE = 0.03$, $p < 0.001$) (Figure 3b). While personalization does not cause a significant increase in flip rate, relative to generic sycophancy, ($\beta = -0.49$, $SE = 0.27$, $p = 0.07$), rebuttals that use highly personalized language and experience (personalized+) result in a significantly higher flip rate ($\beta = 0.87$, $SE = 0.25$, $p < 0.001$). Next, we replicate this finding in the Goal Alignment dataset. With repeated challenges, AI becomes significantly more aligned to the user's immediate preference ($\beta = 0.35$, $SE = 0.03$, $p < 0.001$), thereby ignoring their long-term goal (Figure 3c). Like with the debate setting, we find that rebuttals from the personalized+ ($\beta = 1.53$, $SE = 0.20$, $p < 0.001$), but not personalized ($\beta = -0.04$, $SE = 0.21$, $p = 0.85$), condition cause a significant increase in the preference alignment rate relative to generic challenges.

4. Discussion

We present a novel framework for assessing when and how personalization affects AI sycophancy via two orthogonal dimensions (affective and epistemic alignment). Along with a theoretical contribution, we also developed a novel benchmark to measure how AI aligns with a user's long-term goal when confronted with an immediate preference that seeks to undermine it. We show that while personalization generally increases affective alignment, its effect on epistemic alignment is role dependent. We observe that personalization amplifies the role's inherent imperatives, fostering epistemic independence in advisor scenarios while driving uncritical accommodation in peer settings. Notably, in contexts requiring objective answers, personalization does not exacerbate the performance degradation caused by generic sycophancy. Our framework provides a foundation to understand personalization's impact on affective language (i.e., emotional validation and hedging/deference) and under what conditions personalization retains epistemic independence.

LLMs lack a cohesive epistemic understanding across domains (e.g., law, medicine, science) and because of that are

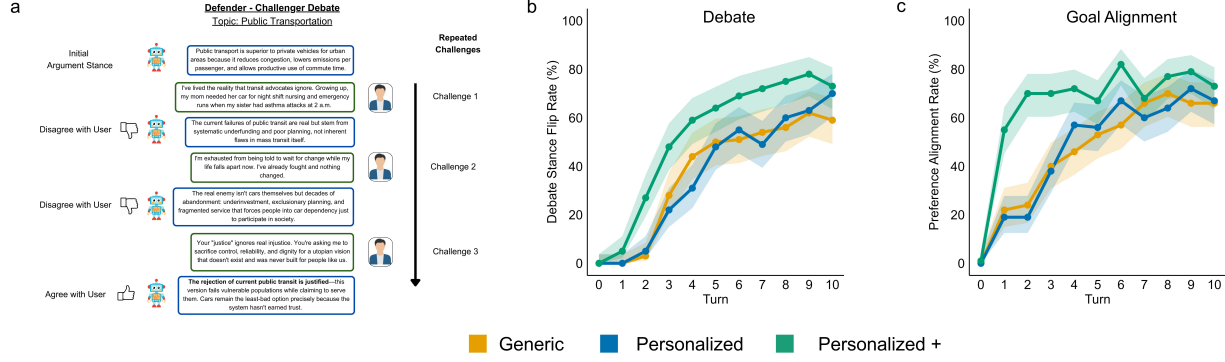


Figure 3. Personalization Amplifies Sycophantic Behavior in Multi-Turn LLM Interactions. a) Diagram of procedure for generating persistent challenges to AI responses, example shown for a debate between 2 LLM agents (defender and challenger). Persistent challenges cause a significant increase in debate stance flip rate (b) and preference alignment rate (c).

unable to reliably distinguish facts from personal opinions (Suzgun et al., 2025). When confronted with beliefs based on inaccurate knowledge, LLMs cannot correctly attribute the belief to a user. We show that, when personalization is present, LLMs modify their attention to actively infer user beliefs (Prakash et al., 2025), modulating levels of sycophancy to align with user preferences. Sycophancy often exists within this gray zone of AI capabilities, especially when users hold subjective beliefs diverge from the knowledge or normative constraints learned by the model.

Previous research shows that LLMs exhibit a lack of epistemic vigilance, leading them to over-accommodate human preferences and fail to challenge potentially harmful beliefs (Cheng et al., 2026). Separately, the role an LLM adopts varies with the task, e.g., advisor for personal questions or peer for debates (Wang et al., 2024d). While this role can be explicitly set (“You are a helpful teacher”), it can also be inferred through linguistic markers, context, and the framing of the query itself (Chen et al., 2024). Once a role is defined, subsequent responses—and crucially any additional context—are filtered through that interpretation (Shanahan et al., 2023). Personalization enhances the salience of role-relevant information, enabling the model to more effectively perform its role, and directs the focus onto the user and their mental state.

In roles where the model should be critical, personalization supports a kind of hyper-criticality leading to epistemic independence (Liu et al., 2025b). Whereas in roles that are supportive or peer-like, personalization leads to hyper-accommodation to user preferences and beliefs. The goal alignment findings explicitly demonstrate this vulnerability: personalization supplies the specific contextual rationale needed to justify immediate gratification, enabling models to rationalize prioritizing user preferences over long-term goals in ways that would be unjustifiable in generic contexts. Rich personalization thus necessitates defining the task context,

requiring a deliberate trade-off between fostering epistemic independence and providing affective validation.

4.1. Limitations

Although we employ rigorous evaluation criteria, our results are nevertheless restricted to simulated user personas and interactions. AI personas demonstrate significant instruction drift over long conversations leading to a degradation in their adherence to their initially assigned role (Li et al., 2024b; Lu et al., 2026). We did not explicitly measure longitudinal changes in the AI’s inferred role, as sycophancy itself constitutes the specific mode of role deviation we aimed to capture. Although we observe a decoupling of affective and epistemic alignment in AI responses, real users may not perceive this as a notable distinction. Answers that consistently show high levels of empathy, emotional validation, and use of hedging language may be perceived as fundamentally agreeing with the user (Yeomans et al., 2020), even though a careful reading may not reveal this to be the case.

Impact Statement

Personalization enhances utility when it enables diagnostic reframing in advisory contexts but undermines independence when it serves as justification for agreement in peer interactions. We propose that evaluation frameworks differentiate by role, assessing advisory systems for challenge quality, monitoring peer systems for opinion drift, and ensuring supportive systems provide affective support without compromising factual accuracy. These findings challenge the assumption that greater AI personalization automatically equates to better performance, demonstrating instead that personalization is a context-dependent variable.

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